C PROGRAMMING — ASSIGNMENT 1

Task 8: Trace the Compilation Process

Marks: 4 | Total assignment: 50 marks

Compilation stages

For a source file task8.c, run: gcc -E task8.c -o task8.i (preprocessed C); gcc -S task8.c -o task8.s (assembly); gcc -c task8.c -o task8.o (object file); gcc task8.o -o task8 (linked executable). The preprocessor expands macros and includes headers and processes conditional compilation. The compiler translates C into assembly; the assembler creates an object file; the linker resolves external references and creates an executable.

Errors and warnings

A compile-time error includes a syntax error such as a missing semicolon. A link-time error occurs if a declared function is called but no definition is linked. A run-time error can be invalid input or an attempted division by zero. gcc -Wall -Wextra helps identify suspicious conversions, unused variables and format mismatches; fix warnings rather than suppressing them.

Multi-file build

With input.c, tariff.c and report.c plus a shared header, compile each source separately to object files, then link them. After changing one .c file, only that object needs recompiling, plus relinking; a single-file program generally recompiles the whole translation unit. Object files consume extra disk space but improve modularity and incremental build time.

C Source Code / Implementation

```
/* task8.c: use this small file to inspect compilation stages */
#include <stdio.h>
#define RATE_PAISE 225
static long charge(int units) {
    return (long)units * RATE_PAISE;
}
int main(void) {
    printf("Charge for 10 units = %ld paise\n", charge(10));
    return 0;
}
/* Commands:
   gcc -Wall -Wextra -E task8.c -o task8.i
   gcc -Wall -Wextra -S task8.c -o task8.s
   gcc -Wall -Wextra -c task8.c -o task8.o
   gcc task8.o -o task8
   ./task8
*/
```

Submission note

Compile and run the code on your own system, save actual terminal output/screenshots, and include them in your report. Review the code and be prepared to explain it during the viva. Any machine-specific measurements must come from your own execution.